

SCARAB workflow

1. Sequence and read inputs

Assembly + FASTQs + optional trusted genomes

One metagenome assembly; tab-separated read list; optional SAG/reference FASTAs.



2. Sequence windows and anchors

Subcontigs + sourmash MinHash

Length filtering and overlapping windows; trusted sequence matches where available.



3. Abundance and composition

Read coverage + tetranucleotide frequencies

minimap2 → SAMtools → MetaBAT depth summary; standardized feature matrices.



4. Parameter selection

Abundance entropy and AutoOpt

Bundled reference tables select algorithm defaults or calibrated parameter sets.



5. Recruitment and clustering

UMAP, HDBSCAN, and one-class SVM

De novo bins; trusted-anchor recruitment; combination/intersection of evidence.



6. Contig and xPG products

Original-contig FASTAs + extended partial genomes

Compile assignments; combine trusted sequence with recruits; BBTools deduplication.



7. Review and downstream analysis

Logs, assignment tables, and FASTA outputs

Validate genome quality independently; retain input and parameter provenance.